Q- 01: What is a virus, virion, prion, and viroid? Give the characteristics and classification of viruses.

Q- 02: Draw and label the structure of a naked and enveloped virus. Write the difference between a virus and bacteria. Give the difference between a naked and enveloped virus.

Q- 03: Write down the classification and characteristics of the Herpes virus. Name the herpes viruses and the diseases produced by them.

Q- 04: Write a short note on Viral Latency.

Q- 05: Describe viral replication with a diagram.

Q- 06: Name the tests available for the diagnosis of a virus in the lab. Discuss the approaches for the diagnosis of viral infection from a specimen.

Q- 07: What is an arbovirus? Name some arboviruses. Give the characteristics of the dengue virus. Describe the clinical types and lab diagnosis of dengue infection. How can one take preventative measures against dengue? Differentiate between classical dengue and DHF in the lab.

Q- 08: What are antigenic shift and antigenic drift? Mention the complications of measles and mumps viral infections. Classify orthomyxoviruses. Mention the difference between orthomyxovirus and paramyxovirus.

Q- 09: Write a short note on viral vaccines.

Q- 10: Write a short note on herd immunity.

Q- 11: Write a short note on SARS-CoV-2.

Q- 12: Give the properties, transmission, pathogenesis, clinical features, and lab diagnosis of the rabies virus. How would you manage a person bitten by a rabid dog?

Q- 13: Define HIV infection and AIDS. Why is HIV called a retrovirus? How is HIV transmitted? List 4 opportunistic infections that occur in an AIDS patient. Write down the lab diagnosis of HIV infection in an adult and a newborn baby of an HIV-infected mother. What are the tests done to evaluate the prognosis of HIV infection?

Q- 14: Name the viruses that are transmitted through the placenta. Name five oncogenic viruses and five helical viruses, and the cancers caused by them.

Q1. What is a virus, virion, prion, and viroid? Give the characteristics and classification of a virus.

Virus: Viruses are the smallest obligate intracellular parasites, having only one kind of nucleic acid in their genome, either DNA or RNA, and cannot replicate independently outside the host cell.

Virion: A complete virus particle that is infectious is called a virion.

Prion: It is an infectious particle that is composed solely of protein, but there is no detectable nucleic acid present.

- e.g., Kuru in humans
- Mad cow disease
- Scrapie disease of snakes.

Viroid: It contains a single molecule of circular RNA without an envelope. It causes several plant diseases but does not cause any human disease.

Pseudoviroid: It contains host cell DNA instead of viral DNA within the capsid.

Defective virus: Composed of viral nucleic acid and protein but cannot replicate without a helper virus. e.g., Hepatitis D (needs HBV).

Characteristics of a Virus:

1. Viruses are not cells, they are the smallest infectious agents.
2. Composed of an internal core containing either DNA or RNA, covered by a protective protein coat.
3. Some viruses have an outer lipoprotein membrane, called an envelope.

Classification of Viruses:

Based primarily on the nature of the genome and whether the virus has an envelope.

A. DNA Viruses

1. **Enveloped Viruses**
    - Hepatitis B virus (HBV) → Hepatitis
    - Herpes simplex virus (HSV) → Conjunctivitis, encephalitis
    - Cytomegalovirus (CMV) → Encephalitis

2. **Non-enveloped/Naked Viruses**
    - Adenovirus → Adenoviral common cold
    - Human papillomavirus (HPV) → Skin lesion
    - BK virus → BKN
    - B19 parvovirus → Slapped-cheek disease

B. RNA Viruses

1. **Enveloped Viruses**
    - Measles virus → Measles
    - Mumps virus → Mumps
    - Respiratory syncytial virus (RSV) → Bronchiolitis
    - Influenza virus → Pneumonia
    - Rubella virus → Rubella
    - Rabies virus → Rabies
    - HIV → AIDS
    - Corona virus → SARS, Common cold
    - Hepatitis C virus → Hepatitis
    - Dengue virus → Dengue
    - Yellow fever virus → Yellow fever

2. **Non-enveloped/Naked Viruses**
    - Hepatitis A virus → Hepatitis
    - Hepatitis E virus → Hepatitis
    - Polio virus → Polio
    - Rhinovirus → Common cold
    - Rotavirus → Diarrhea

Q2. Draw and label the structure of a naked and enveloped virus. Write the difference between virus and bacteria. Give the difference between naked and enveloped virus.

Basic Structure of a Virus:

1. **Genome or Nucleic Acid Core:**

   (a) Single or double-stranded DNA/RNA.

   (b) Codes for the synthesis of viral components and enzymes for replication.

   Importance: Detection of nucleic acid in the diagnosis of viral disease.

2. **Capsid:**

   (a) The nucleic acid is surrounded by a protein coat called a capsid.

   (b) The structure composed of the nucleic acid genome and the capsid protein is called a nucleocapsid.

   Important Functions of Capsid:

   - (a) Protects the genetic material.
   - (b) Mediates the attachment of the virus to specific receptors on the host cell surface.
   - (c) Important antigens.

3. **Envelope:** The envelope is a lipoprotein membrane composed of lipids derived from the host cell membrane and protein that is virus-specific.
   - **Importance:**
     - (a) Fusion of the virus with the host cell membrane.
     - (b) Sensitive to heat, detergent, and lipid solvents such as alcohol and ether.
     - (c) Produces neutralizing antibodies which are important for the diagnosis of the disease and immune status.

Difference between Virus and Bacteria:

Traits	Virus	Bacteria
1. Cell	No	Yes (Prokaryotic)
2. Size	Very small	Larger

3. Outer surface	Protein	Rigid wall
4. Capsid & envelope	Present	Absent
5. Type of nucleus	None	Prokaryotic
6. Nucleic acid	Either DNA or RNA	Both DNA and RNA
7. Ribosome	Absent	Present
8. Multiplication	Replication	Binary fission
9. Cultivation	Cell line	Artificial culture media
10. Visibility	Not visible under light microscope	Visible under light microscope
11. Antibiotic sensitivity	Not sensitive	Sensitive

Difference between Naked and Enveloped Virus:

Traits	Enveloped Virus	Naked Virus
1. Definition	Has a lipid membrane outside the protein capsid.	Lacks a lipid membrane surrounding the protein capsid.
2. Components	Envelope and nucleocapsid.	Nucleocapsid only.
3. Virulence	Less virulent.	More virulent.
4. Sensitivity	Sensitive to dryness, acid, or heat.	Resistant to heat, dryness, or acid.
5. Host cell lysis	Very rare.	Often.
6. Survival in the GI tract	Cannot survive.	Can survive.
7. Example	HIV, HBV, CMV	Adenovirus, HAV, HEV

Q3. Write down the classification and characteristics of Herpes virus. Name the herpes viruses with diseases produced by them.

Classification of Herpes Virus:

Based on the type of cell infected and the site of latency, herpes viruses are of three types:

1. **Alpha herpes virus:** They infect epithelial cells primarily and cause latent infection in neurons. They are fast-growing.
    - **Example:** Herpes simplex virus 1 & 2, Varicella-zoster virus.
2. **Beta herpes virus:** They become latent in secretory glands and kidneys. They are slow-growing.
    - **Example:** Cytomegalovirus, Herpes virus 6 & 7.
3. **Gamma herpes virus:** They infect and become latent in lymphoid cells.
    - **Example:** Epstein-Barr virus, Human Herpes virus 8.

Important Properties of Herpes Virus:

1. They have an icosahedral core surrounded by a lipoprotein envelope.
2. The genome is linear double-stranded DNA.
3. They replicate in the nucleus.
4. They are the only viruses that obtain their envelope by budding from the nuclear membrane.
5. The virion possesses a tegument located between the nucleocapsid and envelope.
6. They have the ability to cause latent infection.
7. They produce a vesicular rash in both primary infection and reactivation (HSV-1 & 2, Varicella-zoster virus).
8. They cause the formation of multinucleated giant cells (HSV-1 & 2, Varicella-zoster virus, CMV).
9. Some are cancer-causing. Example: Epstein-Barr virus, Human Herpes virus-8.
10. It is the second-largest virus (the largest is Poxvirus).

Diseases Produced by Herpes Viruses:

1. **Herpes Simplex Virus-1 (HSV-1):** The lesion is above the waist, and transmission occurs by saliva.
    - Gingivostomatitis
    - Herpes labialis
    - Keratoconjunctivitis
    - Encephalitis
    - Herpes whitlow

2. **Herpes Simplex Virus-2 (HSV-2):** The lesion is below the waist, and transmission occurs by sexual contact.
    - Genital herpes
    - Neonatal herpes
    - Aseptic meningitis

3. **Varicella-Zoster Virus (VZV):**
    - Chickenpox (Varicella)
    - Shingles (Zoster)
4. **Cytomegalovirus (CMV):**
    - **In neonate:** Cytomegalic inclusion disease. Congenital diseases include seizures, deafness, jaundice, mental retardation, hepatosplenomegaly, microcephaly, thrombocytopenia. The mortality rate is 20%.
    - **In immunocompromised host:** Interstitial pneumonitis (bone marrow transplant), bronchiolitis (lung transplant), graft atherosclerosis (heart transplant).
    - **In normal host:** Heterophil antibody-negative infectious mononucleosis-like disease.
5. **Epstein-Barr Virus (EBV):**
    - Nasopharyngeal carcinoma
    - Hodgkin and non-Hodgkin lymphoma
6. **Human Herpes Virus-8 (HHV-8):**
    - Kaposi's sarcoma

Q4. Short note - Viral Latency

The acute disease is followed by an asymptomatic period during which the virus remains in a quiescent state. However, when the patient is exposed to a stimulating agent (sunlight, fever, stress) or has immunosuppression, reactivation of viral replication and disease can occur.

Example Virus	Latency Site
Herpes simplex virus-1	Cranial sensory ganglia
Herpes simplex virus-2	Lumbar or sacral sensory ganglia
Varicella zoster virus	Cranial or thoracic sensory ganglia
Epstein-Barr virus	B lymphocytes
Cytomegalovirus	Uncertain, may be latent within circulatory lymphoid cells or epithelial cells

Q5. Describe viral replication with a diagram.

Replication: Viral replication is the formation of biological viruses during the infection process in the target host cell.

Stages of Viral Replication:

Attachment and penetration by parental virion

↓

Uncoating of the viral genome

↓

Early viral mRNA synthesis

↓

Early viral protein synthesis

↓

Viral genome replication

↓

Late viral mRNA synthesis

↓

Late viral protein synthesis

↓

Progeny virion assembly

↓

Virion release from the cell

Detailed Steps:

- **Attachment/Adsorption:**

    1. Attachment is the interaction of a virion with a specific receptor site on the surface of a cell.
    2. Receptors are generally glycoproteins.
    3. Not all cells are susceptible to a particular viral infection due to a lack of necessary receptors (host cell receptor, organ-specific receptor).

- **Penetration:**

    1. In this stage, the virus particle is taken up inside the cell.
    2. In some systems, it is done by receptor-mediated endocytosis.
    3. Some viruses penetrate directly across the plasma membrane.

4. In some cases, there is a fusion of the virion envelope with the plasma membrane.

- **Uncoating:**

    1. Uncoating is the physical separation of the viral nucleic acid from the outer structural components of the virion so that it can function.
    2. It occurs concomitantly with or shortly after penetration.
    3. Cell enzymes from lysosomes strip off the virus protein coat.
    4. It requires acidic pH in the endosome.
    5. The genome may be released as free nucleic acid or as a nucleocapsid.

- **Viral Messenger RNA Synthesis:**

    1. Specific mRNA is transcribed from the viral nucleic acid.
    2. mRNA is used to instruct the host cell to make viral components.
    3. After this is accomplished, viruses use host cell components to translate the mRNA.
    4. Various classes of viruses use different pathways to synthesize the mRNA depending on the structure of the viral nucleic acid.

- **Synthesis of Virus Components:**

    1. Virus mRNA is translated on host cell ribosomes into two types of viral protein:
        - (a) Structural
        - (b) Non-structural (mainly enzymes for virus genome replication)
    2. Nucleic acids are synthesized by using templates that are either the parental genome or, in the case of single-stranded genomes, newly formed complementary strands.

- **Virion Assembly:**

    1. Newly synthesized viral genomes and capsid polypeptides assemble together to form progeny virus.
    2. This may take place in the cell's nucleus or cytoplasm.

- **Release:**

    1. Non-enveloped viruses are released by lysis of the infected cells.
    2. Enveloped viruses are released by a budding process.

Q6. Name the tests available for diagnosis of virus in the lab. Discuss the approaches for diagnosis of viral infection from a specimen.

Names of Tests Available for Diagnosis of Virus:

A. Identification of viruses using cell culture.

B. Microscopic examination/identification directly in the specimen.

C. Serological procedure to detect a rise in antibody titre or the presence of IgM antibody.

D. Detection of viral antigens in blood or body fluid.

E. Detection of viral nucleic acid in blood or the patient's cells.

Cultivation of Virus:
A. In embryonated eggs:

1. Duck embryo culture
2. Chicken embryo culture

Application: Diagnostic purpose, Vaccine preparation.

B. In cell culture: Means cells grown in a culture media in the lab.
Type of cell culture:

1. Primary culture
2. Diploid cell line culture
3. Continuous cell line culture

Laboratory Diagnosis of Virus:

A. Specimen Collection: Skin scraping from the lesion.

B. Identification of viruses in cell culture.

C. Viral antigen detection (Capsid protein): ELISA / RIA.

D. Nucleic acid detection by PCR.

E. Haemagglutination assay.

F. Electron microscopic examination.

A definitive identification of the virus grown in cell culture is made by using known antibody in one of several tests:

1. Complement fixation test
2. Haemagglutination inhibition
3. Neutralization of the cytopathic effect (CPE) are the most frequently used tests. Other procedures such as fluorescent antibody, RIA, ELISA, and immuno-electron microscopy are also used in special instances.

If the virus does not produce a cytopathic effect, its presence can be detected by several other techniques:

1. Hemadsorption.
2. Interference with the formation of a CPE by a second virus.

Haemagglutination / Hemadsorption Assay:

1. Certain viruses contain specific proteins called haemagglutinins that have the ability to agglutinate RBCs.
2. Attachment of RBC to the surface of virus-infected cells. This technique is limited to viruses with a haemagglutinin protein on their envelope, such as Mumps, Influenza, and Parainfluenzae.

Haemagglutination Inhibition:

1. Many viruses clump red cells from one species or another. This can be inhibited by an antibody specifically directed against the virus.
2. If the virus and antibody are homologous, the virus is blocked from attaching to the erythrocytes, and no haemagglutination occurs.

Interference with the formation of a CPE by a second virus:

For example, the Rubella virus, which does not cause a CPE, can be detected by interference with the formation of a CPE by certain enteroviruses, such as Echovirus and Coxsackievirus.

Microscopic Examination / Identification: 3 different procedures can be used:

1. Light microscopy can reveal characteristic inclusion bodies or multinucleated giant cells.
2. UV microscopy is used for fluorescent antibody staining of the virus in infected cells.
3. Electron microscopy detects virus particles, which can be characterized by their size and morphology.

Serologic Procedure:

1. A serum sample is obtained as soon as a viral etiology is suspected (acute phase), and a second sample is obtained 10-14 days later (convalescent phase).
2. The presence of IgM can be used to diagnose a current infection.
3. The presence of IgG can't be used to diagnose a current infection because the antibody may be due to an infection in the past.

Important properties of HBV:

1. It is a double-stranded DNA virus.
2. The name of the virus family is Hepadnaviridae.
3. It can be transmitted through sexual contact and blood transmission.
4. The envelope contains a protein called the surface antigen (HBsAg), which is important for lab diagnosis and immunization.
5. Within the core is a DNA-dependent DNA polymerase.
6. The DNA polymerase has both RNA-dependent (reverse transcriptase) and DNA-dependent activity.
7. There are two other important antigens: the core antigen (HBcAg) and the e antigen (HBeAg).
8. The core antigen forms the nucleocapsid core of the virion.
9. The e antigen is secreted from infected cells into the blood.
10. The e antigen is an important indicator of transmissibility.

Clinical Features:

1. Anorexia
2. Vomiting
3. Abdominal pain
4. Dark urine

5. Pale feces
6. Jaundice
7. Elevated transaminase level.

Major Complications of HBV:

1. Chronic hepatitis
2. Cirrhosis of the liver
3. Hepatocellular carcinoma

Serological Markers and their Significance:

1. **HBsAg:** Antigen found on the surface of HBV, positive during acute disease. Continued presence indicates a carrier state.
2. **HBsAb:** Antibody to HBsAg. Provides immunity to hepatitis B virus.
3. **HBcAg:** Antigen associated with the core of HBV.
4. **HBcAb:** Antibody to HBcAg. Positive during the window phase. IgM HBcAb is an indicator of recent disease. IgG HBcAb is an indicator of past infection.
5. **HBeAg:** Important indicator of transmissibility.
6. **HBeAb:** Antibody to e antigen. Indicates low transmissibility.

Other viruses that cause hepatitis:

1. Epstein-Barr virus
2. Cytomegalovirus (CMV)
3. Herpes Simplex virus (HSV)
4. Yellow fever virus

Oncogenic viruses:

1. Human T-cell lymphotropic virus
2. Human papillomavirus
3. Epstein-Barr virus
4. HBV
5. HCV
6. Human herpes virus 8

Q 7. What is an arbo virus? Name some arbo viruses. Give the characteristics of dengue virus. Describe the clinical types and lab diagnosis of dengue infection. How to take prevention against dengue? Differentiate classical & DHF in the lab.

Zoonotic virus: These are viruses that cause infection common in animal populations that occasionally infect humans.

A. Arthropod-borne virus (Arbovirus)

B. Rodent-borne virus (Robovirus)

- **Example:** Hanta virus, Nipah virus

Arbo virus: It is an acronym for arthropod-borne virus. These viruses are transmitted by arthropods, primarily mosquitoes and ticks.

Members of Arbovirus:

1. Dengue virus
2. Yellow fever virus
3. Chikungunya virus (Toga virus)
4. West Nile virus
5. St. Louis encephalitis virus and Japanese encephalitis virus
6. Eastern equine encephalitis virus
7. Western equine encephalitis virus
8. California encephalitis virus
9. Colorado tick fever virus

Important characteristics of dengue virus:

1. Single-stranded RNA virus.
2. Icosahedral nucleocapsid.
3. Enveloped virus.
4. Four serotypes: DEN-1 to DEN-4.
5. Three structural proteins:
    - C (Capsid protein)
    - prM/M (Membrane glycoprotein)
    - E (Envelope protein)
6. Seven non-structural proteins: NS1, NS2a, NS2b, NS3, NS4a, NS4b, NS5.

Dengue: It is a mosquito-borne infection, characterized by high fever, muscle and joint pain (breakbone fever), rash, lymphadenopathy, and pain on eye movement.

Dengue viral infection types:

1. **Asymptomatic**
2. **Symptomatic**

(a) Undifferentiated fever

(b) Dengue fever

(i) Without hemorrhage

(ii) With hemorrhage

(c) Dengue hemorrhagic fever (DHF)

(i) No shock

(ii) Dengue shock syndrome (DSS)

Pathogenesis of dengue fever:

1. **Agent:** Dengue virus
2. **Host:** Human
3. **Clinical diseases:** Classical dengue fever, Dengue hemorrhagic fever (DHF), Dengue shock syndrome (DSS)
4. **Reservoir of infection:** Human
5. **Vector:** Female Aedes aegypti
6. **Mode of transmission:** By the bite of an infected Aedes aegypti mosquito.

Lab diagnosis of dengue:

1. **Sample:** Blood (in coagulant activator tube)
2. **Detection of antigen:** NS1 antigen detection in the early stage.
3. **Serologic test:** Demonstration of the presence of antibodies.
   - IgM - for primary infection
   - IgG - for secondary infection
   - OR a four-fold or greater rise in antibody titre in acute and convalescent sera.
4. **Methods:** ELISA, ICT, and Haemagglutination inhibition test.
5. **PCR:** Rapid identification and subtyping.
6. **Blood examination:** Leukopenia, Thrombocytopenia, Increased PCV, Increased SGPT.

Prevention of dengue:

1. No antiviral or vaccine therapy is available for dengue.
2. Outbreaks are controlled by using insecticides and draining stagnant water that serves as the breeding place for mosquitoes.

3. Personal protection includes using mosquito repellent and wearing clothing that covers the entire body.

Description of classical dengue fever:

1. Sudden onset of fever, malaise, pain in the back, joint, muscle, and eyeball, rash, and lymphadenopathy.
2. Fever subsides on the 3rd day and rises again on the 5th to 6th day. Myalgia and deep bone pain (break-bone fever) are characteristic.
3. Viremia is present at the time of fever. Classical dengue fever is a self-limited disease.
4. Produces antibody (IgG) only for that specific serotype.

Description of dengue hemorrhagic fever and DSS:

1. When the patient is infected with another serotype of dengue virus.
2. Cross-reacting antibodies produced against the first serotype.
3. Immune complex deposition.
4. Activation of the classical complement pathway.
5. Increased vascular permeability (C3a, C5a), local hemorrhage, loss of blood/plasma leakage -> hypovolemia -> shock.
6. Increased vascular permeability by IL-2.
7. Abnormal hemostasis due to vasculopathy, thrombocytopenia, and coagulopathy.
8. It is a much more severe disease with a fatality rate.

Q 8. What is antigenic shift and antigenic drift? Mention the complication of measles and mumps viral infections. Classify orthomyxovirus. Mention the difference between orthomyxovirus & paramyxovirus.

Classification of Orthomyxovirus:

1. **Influenza A virus:** Causes worldwide epidemics of influenza.
2. **Influenza B virus:** Causes major outbreaks of influenza.
3. **Influenza C virus:** Causes mild respiratory tract infections.

Diagram Description: A diagram showing reassortment between a human influenza virus and a chicken influenza virus within a lung cell, resulting in a new strain of influenza virus. This process is labeled "Antigenic Shift".

Difference between Orthomyxovirus and Paramyxovirus:

Property	Orthomyxovirus	Paramyxovirus

1. Viruses	Influenza A, B, C	Measles, Mumps, RSV, and parainfluenza virus
2. Genome	Segmented (8 pieces) ssRNA of negative polarity	Non-segmented ssRNA of negative polarity
3. Surface spikes	Hemagglutinin and neuraminidase on different spikes	Hemagglutinin and neuraminidase on the same spike
4. Giant cell formation	No	Yes

Antigenic shift: Major changes based on the reassortment of segments of the genome RNA. Reassortment means entire segments of RNA are exchanged, each of which codes for a single protein (e.g., hemagglutinin). This change appears about every 10 years and mainly in influenza A virus.

Antigenic drift: Minor changes based on mutations in the genome RNA. This change appears about every year and mainly occurs in influenza B and C viruses.

Measles Virus:

Disease produced is measles, characterized by a maculopapular rash. It occurs primarily in childhood.
Transmission: Transmitted via respiratory droplets produced by coughing and sneezing.

- **Clinical features of measles:** Incubation period: 10-14 days.
    - Fever, conjunctivitis, running nose, and coughing occurs.
    - Koplik's spots - bright red lesions with a white, central dot that are located on the buccal mucosa and are virtually diagnostic.
    - Maculopapular rash appears on the face and proceeds gradually down the body to the lower extremities, including the palms and soles
- **Complication of measles:**
    - **Acute complications:**
        1. Otitis media
        2. Diarrhea
        3. Pneumonia (Giant cell pneumonia)
        4. Acute encephalitis
    - **Chronic complications:**
        1. Post-infectious encephalomyelitis (PIE)
        2. Subacute sclerosing panencephalitis (SSPE)

3. Measles inclusion body encephalitis (MIBE)

Mumps Virus:

Disease is mumps, characterized by parotid gland swelling. It occurs primarily in childhood. **Transmission:** Transmitted via respiratory droplets. Mumps occurs worldwide with a peak incidence in the winter. About 30% of children have a subclinical infection which confers immunity.

- **Clinical features:**

    1. Incubation period: 18-21 days.
    2. A prodromal stage: fever, malaise, anorexia, tender swelling of the parotid glands either unilateral or bilateral.
    3. There is a characteristic increase in parotid pain when drinking citrus fluid.
    4. This disease is typically benign and resolves spontaneously within 1 week.

- **Complication:**

    1. Acute meningitis (15%)
    2. Meningoencephalitis
    3. Deafness (unilateral)
    4. Pancreatitis (4%)
    5. Orchitis (unilateral and bilateral, can cause sterility)
    6. Oophoritis in females.

Q10. Short note on viral vaccine.

Prevention of viral disease can be achieved by the use of vaccines that induce active immunity or by the administration of a preformed antibody that provides passive immunity.

Active Immunity: Three types of vaccines that induce active immunity:

1. Live vaccine
2. Killed vaccine
3. Subunit vaccine

Live vaccine: Those that contain live viruses whose pathogenicity has been attenuated.

- **Example:** Polio vaccine (oral), MMR vaccine, Influenza vaccine, Varicella (chickenpox) vaccine, Rota virus vaccine.

Live viral vaccines are preferable to killed vaccines for three reasons:

1. Induce a higher titre of antibody and hence longer-lasting protection.
2. Induce a broader range of antibodies (e.g., IgA and IgG).
3. Activate cytotoxic T cells which kill virus-infected cells.

Advantages of live vaccine:

1. Longer duration of protection.
2. More protective.
3. Induce IgG and IgA antibodies.
4. Induce cell-mediated immunity (CMI).
5. Produce herd immunity (if it is excreted by the immunized person).

Disadvantages of live vaccine:

1. Should not be given to immunocompromised individuals and pregnant women.
2. Can revert to virulence.
3. Less heat stable.

Killed vaccine: Those that contain killed virus.

Example: Polio (IPV - Injectable polio vaccine), Influenza, Rabies, Hepatitis A.

Advantages of killed vaccine:

1. Can be given to immunocompromised individuals and pregnant women.
2. Cannot revert to virulence.
3. More heat stable.

Disadvantages of killed vaccine:

1. Shorter duration of protection.
2. Less protective.
3. Induce only IgG antibodies.

Subunit vaccine: Contains purified viral proteins.

Hepatitis B vaccine, Human papillomavirus vaccine.

Passive Immunity:

- **Advantages:** Provides immediate protection.
- **Disadvantages:** Does not provide long-term protection. It is active only for a few weeks to a few months.
    - **Example:** Rabies vaccine, Hepatitis A vaccine, Hepatitis B vaccine, Measles vaccine, Varicella-zoster vaccine.

Passive-Active Immunity: It consists of administering both immunoglobulin and a viral vaccine. This provides both immediate as well as long-term protection.

- **Example:** Rabies in an unimmunized person, Hepatitis B vaccine.

Q11. Short note on herd immunity.

Herd immunity is the protection of an individual that results from immunity in many other members of the population, which interrupts the transmission of the virus to the individuals. Herd immunity can be achieved either by active immunization or by natural infection of a sufficiently high percentage of the population. Active immunization - live polio vaccine.
The live vaccine can be excreted by the immunized person. This is a double-edged sword. It is advantageous if the spread of the virus successfully immunizes others, as occurs with the live polio vaccine.

Q12. Short note on SARS-CoV-2.

Important properties: Non-segmented single-stranded, positive-polarity RNA genome. Enveloped virus with a helical nucleocapsid. In the electron microscope, prominent club-shaped spikes in the form of a corona can be seen. Consists of 4 structured proteins:

1. Spike (S)
2. Membrane (M)
3. Envelope (E)
4. Nucleocapsid (N)

7 serotypes of human coronaviruses, 4 of which cause URTI like the common cold. 3 cause LRTI like pneumonia. Coronaviruses that cause pneumonia are SARS-CoV, SARS-CoV-2, MERS-CoV. The receptor for SARS-CoV-2 on the human cell is ACE-2 (Angiotensin-Converting Enzyme-2) protein.

- **Transmission:**
    - Inhalation of respiratory droplets by coughing, sneezing.
    - Transmission by hand contact with surfaces containing the virus. Fingers transport the virus on the surface to the recipient's eyes, nose, or mouth.
    - Infections are acquired by transmission from asymptomatic carriers, probably by droplet/aerosols.

- **Pathogenesis:**

    - SARS-CoV-2 binds with spike proteins to the ACE-2 protein and neuropilin-1 on the surface of the respiratory tract epithelium.
    - Transmembrane protease serine 2 (TMPRSS2) cleaves the spike protein to reveal the fusion protein.
    - Fusion of the virus with the cell membrane.
    - Entry of the virus into the cytoplasm of alveoli.
    - Respiratory manifestation is likely to have 2 pathogenic mechanisms:
        - (a) Killing of alveolar cells by the virus -> Accumulation of cell debris blocks diffusion of oxygen into the capillaries -> Hypoxia.
        - (b) Killing of the endothelial cells of the capillaries lining the alveoli -> This triggers blood clots and an immune-mediated "cytokine storm" -> ARDS. Damage to endothelial cells and blood clots can cause many respiratory manifestations.

- **Clinical features:**
    - Incubation period: 2-14 days.
    - Asymptomatic or mild to severe infection.
    - **S/S:** Prolonged and intermittent fever, myalgia/fatigue, sore throat, cough, sputum, shortness of breath, headache, hemoptysis, diarrhea.

- **Diagnosis:**
    - **Specimen:** Nasopharyngeal/oropharyngeal swab, bronchoalveolar lavage, sputum, blood.
    - **Screening for tests (blood):** WBC count, CRP and ESR, D-dimer and PT, ALT/AST, IL-6.
    - **Rapid antigen test (RAT):** which detects the viral antigen protein of SARS-CoV-2.
    - **Nucleic Acid Test (NATs):** by RT-PCR, which detects SARS-CoV-2 viral RNA.
    - **Detection of IgM and/or IgG antibody against SARS-CoV-2 by ELISA.**
    - **Genome sequencing.**
    - **Chest X-ray / Chest CT scan:** Multiple areas of consolidation and ground-glass opacities are typical findings.

- **Important treatment modalities used in COVID-19:**
    - **Therapeutic purpose:**
        - Inhibit viral replication: Remdesivir, Molnupiravir.
        - Inhibit cytokine storm: Dexamethasone.
        - Monoclonal antibody against IL-6 receptor: Tocilizumab.

- **COVID-19 vaccine:**
  - RNA vaccines (mRNA): Pfizer and Moderna.
  - Viral vector vaccine: AstraZeneca, Covishield, Johnson and Johnson.
  - Protein-based vaccine: Novavax.
  - Inactivated virus vaccine: Sinovac, Sinopharm.

Of course, here is the final part of the transcribed notes.

Q13. Give the properties, transmission, pathogenesis, clinical features and lab diagnosis of rabies virus. How would you manage a person bitten by a rabid dog?

Properties of rabies virus:

1. The rhabdoviruses are rod or bullet-shaped particles.
2. The particles are surrounded by a membranous envelope with protruding spikes, composed of trimers of viral glycoprotein.
3. Inside the envelope, there is a nucleocapsid.
4. The genome is single-stranded, negative-sense RNA.
5. The virion contains RNA-dependent RNA polymerase.
6. Rabies virus survives at 4°C for weeks and at -70°C for years.
7. The virus is killed rapidly by exposure to ultraviolet radiation or sunlight, heat (at 50°C for 1 hour), lipid solvents, trypsin, detergents, and by extremes of pH.

Transmission:

1. By the bite of a rabid animal, such as a dog, cat, fox, etc.
2. Licking from a rabid animal, as the virus resides in saliva.
3. Aerosol from wild birds.
4. Transplant of cornea from patients who died of undiagnosed rabies.

Pathogenesis:

1. Virus enters via a rabid animal bite.
2. Virus replicates in muscles at the site of the bite.
3. Virus infects nerves in the peripheral nervous system and moves by retrograde transport.

4. Virus replicates in the dorsal root ganglion and travels up the spinal cord to the brain.
5. Brain becomes infected.
6. Virus travels from the brain via nerves to other tissues such as the eye, kidney, and salivary gland.

Clinical features: Rabies is an acute infection of the central nervous system leading to encephalitis with a fatal outcome.

- **Clinically three phases:**
    1. **Short prodromal phase:** Malaise, headache, anorexia, nausea, vomiting.
    2. **Acute neurologic phase:** Signs of nervous system dysfunction, sympathetic overactivity occurs (increased salivation, perspiration), hydrophobia occurs.
    3. **Coma:** There is hallucination, muscle spasm, convulsion, cranial nerve lesion, and hyperpyrexia. Coma and death occur in 2-5 days, usually due to respiratory paralysis.
-

Lab diagnosis:

1. A rabid antigen or nucleic acid.
2. **Specimen:** Biopsy specimen from the neck at the hairline, impression preparation of the brain and cornea.
3. Tissue infected with rabies is identified by immunofluorescence or immunoperoxidase staining using anti-rabies monoclonal antibodies.
4. A definitive pathogenic diagnosis is made by finding Negri bodies.
5. PCR can be used to amplify parts of the rabies virus genome from fixed or unfixed brain tissue and saliva.
6. **Serology:** Serum antibodies can be detected by immunofluorescence or neutralization tests.
7. **Virus isolation:** The virus can be isolated from animal brain by growth in cell culture, but this takes a long time.

Type of vaccine:

1. Human diploid cell vaccine (HDCV)
2. Purified chick embryo cell vaccine
3. Nerve tissue vaccine
4. Duck embryo vaccine

Management of animal bite:

1. All bites should be thoroughly cleaned with soap and water.
2. Local treatment of the wound.
3. Prompt and adequate local treatment of all bite wounds and scratches is the first requisite.

4. **Cleansing:** Immediate flushing and washing the wound, scratches, and the adjoining area with plenty of soap and running tap water.
5. If soap is not available, at least simple flushing of the wound with plenty of water should be done.
6. **Chemical treatment:** The residual virus after cleansing should be inactivated by irrigation with virucidal agents.
7. **Immediate vaccination:** Intramuscular 5 doses of HDCV (1 ml IM) on days 0, 3, 7, 14, and 28.
8. **Intradermal dose:** 4 doses of rabies vaccine ID on days 0, 3, 7, and 14.
9. Administration of rabies immunoglobulin (RIG) once with the first dose of vaccine.
10. RIG is given as much as possible around the bite site; the remainder is given IM at a different site.
11. Local treatment of the wound - antibiotic administration.
12. Tetanus toxoid.

Q14. Define HIV infection and AIDS. Why is HIV called a retrovirus? How is HIV transmitted? List 4 opportunistic infections that occur in an AIDS patient. Write down the lab diagnosis of HIV infection in an adult and a newborn baby of an HIV-infected mother. What are the tests done to evaluate the prognosis of HIV infection?

HIV: HIV is a virus that causes AIDS. It attacks the immune system of the body. Both HIV-1 and HIV-2 cause AIDS, but HIV-1 is found worldwide, whereas HIV-2 is found primarily in West Africa.
AIDS: Acquired immunodeficiency syndrome is manifested by a decline in the number of CD4+ T-cells to below 200 copies/mm³ of blood and an increase in the severity and frequency of opportunistic infections. With no treatment, the interval between primary infection with HIV and the first appearance of clinical disease is about 10 years.

Important properties:

1. HIV is one of the two important human T-cell lymphotropic retroviruses.
2. HIV infects and kills helper (CD4) T cells, resulting in the loss of cell-mediated immunity.
3. It infects other cells of the immune system (e.g., macrophages, monocytes) that have CD4 protein on their surface.
4. HIV belongs to the Lentivirus subgroup of retroviruses, which causes "slow" infections with a long incubation period.
5. HIV has a bar-shaped core surrounded by an envelope containing virus-specific glycoproteins (gp120, gp41).
6. The genome of HIV consists of two identical molecules of single-stranded positive-polarity RNA.

7. Three structural genes: gag, pol, and env, which encode the internal core, reverse transcriptase, and envelope glycoprotein, respectively.
8. Six regulatory genes: Two of these, tat and rev, are required for replication. The other four, nef, vif, vpr, and vpu, are not required for replication and are termed "accessory" genes.

Why is HIV called a retrovirus? When the virus enters the cell, the RNA is transcribed to DNA by the virally encoded enzyme reverse transcriptase (RT). RT reverses the normal transcription process and makes a DNA copy of the viral RNA genome.

Transmission:

1. Sexual transmission
2. Transfusion of infected blood and blood products
3. Use of injected syringes and needles
4. Perinatal transmission from an infected mother to neonates either through the placenta at birth or via breast milk.
5. Through transplanted organs.
6. Through contaminated instruments.

Opportunistic infections in AIDS:

1. **Bacteria:** *Mycobacterium tuberculosis*, *Mycobacterium avium*, *Nocardia asteroides*, *Salmonella*, *Streptococcus*, *Listeria monocytogenes*.
2. **Protozoa:** *Cryptosporidium*, *Isospora belli*.
3. **Helminths:** *Strongyloides stercoralis*.
4. **Fungi:** *Candida albicans*, *Cryptococcus neoformans*.
5. **Viruses:** Cytomegalovirus, Herpes simplex virus, Hepatitis B virus, Varicella-zoster virus.

Laboratory diagnosis:

1. **Detection of evidence of HIV infection:**
    - (a) Isolation of virus
    - (b) Detection of antiviral antibody
    - (c) Detection of viral antigens and nucleic acid
2.
3. **Detection of immunosuppression:**
4. **Test to detect opportunistic infection or unusual malignancy.**

- **Detection of antiviral antibodies:**
    - Antibodies to gp120, gp41, and p24.
    - **Method:**
        - (a) Screening test: ELISA. (If positive, repeat ELISA).
        - (b) Confirmatory test: Western blot. At least 2 bands have to be positive.
    -

-
- **Detection of viral antigen and nucleic acid:**
    - (a) Detection of p24 antigen by Western blot technique.
    - (b) Detection of viral nucleic acid by RT-PCR.
-
- **Isolation of virus:**
    - (a) Specimen: Peripheral blood
    - (b) Technique: Co-culture
    - (c) Observation: Cytopathic effect (Syncytia and cytolysis)
    - (d) Detecting reverse transcriptase activity by PCR.
    - (e) Detecting virus-specific Ag p24 by immunofluorescence.
-
- **Detection of immune suppression:**
    - TC and DC of WBC - lymphopenia.
    - CD4 T cell count in peripheral blood.
    - **Method:** Flow cytometry and immunofluorescence.
    - **Result:** Reduced count. Reversion of CD4 T cell and CD8 T cell ratio to 1:2.
-

Tests to evaluate prognosis: Tests are done to evaluate the progression of the disease.

Q15. Name the viruses that are transmitted through the placenta. Name five oncogenic viruses and five helical viruses and the cancer caused by them.

Name of the viruses that transmit through the placenta:

1. Cytomegalovirus (CMV)
2. Human immunodeficiency virus (HIV)
3. Rubella virus
4. Zika virus
5. Parvovirus B19

Name of helical viruses:

1. Rabies virus
2. Influenza virus
3. Measles virus
4. Parainfluenza virus
5. Respiratory syncytial virus
6. Mumps virus
7. Corona virus
8. Ebola virus
9. Lymphocytic choriomeningitis virus

10. Lassa fever virus
11. California encephalitis virus
12. Hanta virus

Name of oncogenic viruses:

1. Human T cell lymphotropic virus
2. Human papillomavirus (HPV)
3. Epstein-Barr virus
4. Hepatitis B virus (HBV)
5. Hepatitis C virus (HCV)
6. Human herpes virus-8 (HHV-8)

Immunology

Q1. Define and classify immunity with examples.

Q2. Define antigen, immunogen, epitope, and hapten. Classify antigens. Give the criteria of an immunogen/immunogenicity.

Q3. Name the cells involved in the immune system/innate or acquired immunity. Write down the difference between active and passive immunity. Give the difference between innate and acquired immunity.

Q4. Name the type of antigen-antibody reaction. Define and classify agglutination with examples.

Q5. Name the types of antigen-antibody reaction-based laboratory tests. What is the difference between agglutination and precipitation tests?

Q6. Write down the principle and procedure of ELISA. Give two examples of the ELISA-based test.

Q7. Give the indication, procedure, advice, result, and interpretation of the tuberculin test.

Q8. Define and classify hypersensitivity. Describe the mechanism of anaphylaxis. Give the mechanism of each type of hypersensitivity.

Q9. Short note on opsonization

Q10. Short note on ASO titre

Q11. Short note on T lymphocyte (T-cell).

Q12. List the primary and secondary lymphoid organs. Write down the definition and causes of immune suppression. Define immune response.

Give the difference between primary and secondary immune response. Compare and contrast between B & T lymphocytes.

Q-13. Define immunoglobulin. Give the properties of different immunoglobulins in tabulated form with classification and the difference between IgG & IgM.

Q-14. What is complement? Describe the complement activation pathway.

Q-15. Write a short note on herd immunity, and PCR.

Q-16. Draw and label the structure of an immunoglobulin molecule.

Q-17. Diagram the complement activation pathway. What is the biological function of complement?

Q-18. Write down the classification of immune responses. Differentiate between humoral and cell-mediated immune responses.

Q-19. Give examples of agglutination and precipitation.

Q-20. Write a short note on radioimmunoassay and MHC.

Q-21. Write a short note on the Widal test.

Q-22. Give the properties of IgM, IgA, and IgE.

Q-23. Write a short note on tumor antigens.

Q-24. Write down the comparison between Class I and Class II MHC.

Q-25. Define allergy. Explain Type I hypersensitivity with an example.

Q-26. Write a short note on phagocytosis.

Q-27. Write about CFT (Complement Fixation Test). Mention its use.

Q-28. Principles and interpretation of the Coombs test.

Q-29. Short note on ICT (Immunochromatographic Test).

Q-30. Describe the pathogenesis of anaphylaxis, including the type of antibody, the cells, and mediators involved.

Q-31. What are the causes of secondary immune deficiency?

Q-32. What are the determinants of innate immunity?

Q-33. Classify different types of serological tests performed in the laboratory with their examples. List them properly formatted.

Q1. Define and classify immunity with examples.

Immunity: It is the ability of the human body to recognize, destroy, and eliminate foreign substances which have the tendency to cause tissue damage.

Classification of immunity:

- **A. Innate immunity (from birth)**
- **B. Acquired immunity**
    - **(a) Active (own antibody)**
        - (i) **Natural:** Exposure to an infectious agent (e.g., Hepatitis B).
        - (ii) **Artificial:** Immunization (e.g., Tetanus Toxoid (TT)).
    -
    - **(b) Passive (pre-formed antibodies)**
        - (i) **Natural:** (Immunoglobulins)
            - IgG -> Transplacental transfer from mother to fetus.
            - IgA -> Breast milk through colostrum.
        -
        - (ii) **Artificial:** Acquired from others.
    -
-

Q2. Define antigen, immunogen, epitope, and hapten. Classify antigens. Give the criteria of an immunogen/immunogenicity.

- **Antigen:** Antigens are high molecular substances, usually protein or polysaccharide in nature, and when introduced into suitable individuals through a suitable route, are capable of producing antibodies with which they react specifically.
- **Immunogen:** Immunogens are molecules that induce humoral (B-cell) immunity or cell-mediated (T-cell) immunity, or both.
- **Epitope:** These are small chemical groups on the antigen molecule that can elicit and react with an antibody. They are simply known as the antigen-binding site.
- **Hapten:** These are low molecular substances that in themselves are not immunogenic but when bonded with a carrier protein, they act as an antigen and can produce an antibody.

Classification of antigen:

1. **On the basis of immune response:**
    - (a) Complete antigen (Immunogens)
    - (b) Incomplete antigen (Hapten)
2.
3. **On the basis of antigen-host relationship:**
    - (a) Self or autoantigen
    - (b) Non-self or foreign antigen

        - (i) Alloantigen
        - (ii) Isoantigen (Same species)
        - (iii) Heteroantigen (Different species)
    -
4.
5. **On the basis of the mechanism of antibody formation:**
    - (a) T-dependent antigen
    - (b) T-independent antigen
    - (c) Superantigen
6.

Criteria of an antigen/immunogen:

1. **Foreignness:** Molecules must be recognized as non-self or foreign to the host.
2. **Molecular weight:** High molecular weight, usually more than 10,000 Dalton.
3. **Nature:** Protein or polysaccharide.
4. **Chemical complexity:** Amino acid heteropolymers are more immunogenic.
5. **Route of administration:** Proteins and polysaccharides are immunogenic via the intravenous route.
6. **Antigenic determinants (epitope):** The epitope must be exposed to the immune system.

Q3. Name the cells involved in the immune system/innate or acquired immunity. Write down the difference between active and passive immunity. Give the difference between innate and acquired immunity.

Cells involved in the immune system:

1. **Lymphocytes:**
    - (a) T-Cell
    - (b) B-Cell
    - (c) Natural killer cells
2.
3. **Neutrophils**
4. **Monocytes / Macrophages**

Difference between innate and acquired immunity:

Traits	Innate immunity	Acquired immunity
1. Time of development	Present from birth	Acquired after birth
2. Specificity	Non-specific	Specific

3. Immunological memory	Absent	Present
4. Response to subsequent exposure	Initial and subsequent responses are the same	Initial and subsequent responses are different
5. Cell involvement	Macrophage, polymorphonuclear leukocytes	Predominantly Lymphocyte
6. Present in	Both vertebrates and invertebrates	Only vertebrates
7. Mechanical barrier	Present (skin, mucosa)	Absent

Difference between active and passive immunity:

Traits	Active immunity	Passive immunity
1. Active participation by host	Yes	No
2. Onset of action	Immunity starts to develop after a considerable latent period	Starts immediately
3. Duration of action	Long-lasting	Short
4. Immunological memory	Present	Absent
5. Secondary immune response	Occurs	Not occurs
6. Risk of hypersensitivity	No risk of hypersensitivity	Type-I and Type-II hypersensitivity
7. Example	Natural: exposure to infectious agents -> Hepatitis B. Artificial: Immunization -> Tetanus toxoid (TT)	Natural: IgA, IgG. Artificial: AIDS, ATS etc.

Q4. Name the type of antigen-antibody reaction. Define and classify agglutination with examples.

Type of antigen-antibody reaction:

1. **Direct reactions:**
    - (a) Agglutination reaction
    - (b) Precipitation reaction
2.
3. **Indirect reactions:**
    - (a) Complement fixation test (CFT)
    - (b) Enzyme-Linked Immunosorbent Assay (ELISA)
    - (c) Radio-immunoassay (RIA)
    - (d) Immunochromatographic test (ICT)
    - (e) Immunofluorescence test
4.

Agglutination: When the antigen is particulate and multivalent, the reaction of an antibody with the antigen can be detected by agglutination or clumping of the antigen.

Classification of agglutination:

1. **Direct agglutination:**
    - (a) Slide agglutination: (i) Blood grouping, (ii) Serotyping of bacteria
    - (b) Tube agglutination: (i) Widal test, (ii) Weil-Felix test
2.
3. **Indirect agglutination:**
    - (a) Haemagglutination test: (i) TPHA
    - (b) Latex agglutination test: (i) HBsAg, (ii) ASO titre
    - (c) Co-agglutination test: (i) Antigen detection of *S. pneumoniae*, *N. meningitidis*, *Streptococcus*
    - (d) Particle agglutination test: (i) HIV
4.

Q5. Name the types of antigen-antibody reaction-based laboratory tests. What is the difference between agglutination and precipitation tests?

Antigen-antibody reaction based Laboratory tests:

1. Agglutination test
2. Precipitation test
3. Neutralization test
4. Complement fixation test
5. Enzyme-linked Immunosorbent assay (ELISA)
6. Immunofluorescence test
7. Radio-Immuno assay (RIA)
8. Immunochromatographic test (ICT)

Difference between agglutination and precipitation test:

Agglutination test	Precipitation test
1. Antigen must be a particulate form (e.g., Bacteria, RBC, or coated with latex particle).	1. Antigen must be in soluble form.
2. Done in a small cup, tube, or on a slide.	2. Done in solution or in semi-solid media.
3. Forms clumping or agglutination.	3. Forms aggregates or precipitates and formation of a precipitin curve.
4. A small amount of antibody can be detected.	4. Very few antigens can be detected.
5. Mostly due to IgM.	5. Mostly due to IgG.

Q6. Write down the principle and procedure of ELISA. Give two examples of the ELISA-based test.

ELISA: Enzyme-Linked Immunosorbent Assay. This method can be used for the quantification of either antigen or antibodies in a patient sample.
Principle: It is based on covalently linking an enzyme to a known antibody, reacting the enzyme-linked material with the patient specimen, and then assaying for enzyme activity by adding the substrate of the enzyme.

Types of ELISA:

1. **Indirect ELISA:** Detection of antibody.
2. **Direct / Sandwich ELISA:** Detection of antigen.
3. **Competitive ELISA:** Detection of antigen.

Procedure of ELISA (Indirect ELISA):

1. A sample containing the primary antibody (Ab1) is added to an antigen-coated microtiter well.
2. Incubate to mix up and wash.
3. Free Ab1 is washed off. The presence of an antibody bound to the antigen is detected by adding an enzyme-conjugated secondary antibody (Ab2) that attaches to the primary antibody.
4. Wash again to remove any free Ab2.
5. A substrate for the enzyme is added. The amount of colored reaction product that forms is measured by special spectrophotometric plate readers.

Interpretation:

1. **Production of color:** Indicates the presence of an antibody in the test serum; that is positive.

2. **No color production:** Indicates the absence of an antibody in the test serum; that is negative.

Name of ELISA-based test:

1. **Detection of antibody:**
    - (a) HIV antibody in serum
    - (b) Toxoplasma antibody in serum
    - (c) Kala-azar antibody in serum
2.
3. **Detection of antigen:**
    - (a) Rota virus
    - (b) HBsAg
4.

Q7. Give the indication, procedure, advice, result, and interpretation of the tuberculin test.

Tuberculin test: It is an intradermal skin test that depends on a delayed hypersensitivity reaction which develops in an individual. It becomes positive 4-6 weeks after infection.
Indication:

1. Aid to diagnosis
2. For contact tracing
3. For post-reaction check-up

Procedure: Purified Protein Derivative (PPD) is used as the reagent.

1. 0.1 ml of PPD is slowly injected on the flexor aspect of the forearm intradermally.
2. The test is examined after 72 hours and is carefully palpated for any induration.

Advice:

1. A false positive result may occur due to infection with an atypical mycobacterium.
2. A false negative result may occur due to HIV infection, Measles, Hodgkin's disease, Drugs.

Result:

1. Induration 10 mm or more: Tuberculin test positive.
2. Induration 5 mm or less: Tuberculin test negative.
3. Induration 6-9 mm: Doubtful reaction.

Interpretation:

1. **Positive:** Infected with tuberculosis and/or past infection.

2. **Negative:** Never infected with *Mycobacterium*.
3. **Doubtful:** The test is repeated after 3-4 weeks.

Q8. Define and classify hypersensitivity. Describe the mechanism of anaphylaxis. Give the mechanism of each type of hypersensitivity.

Hypersensitivity: It is an exaggerated, inappropriate immune response to a particular antigenic stimulus in a pre-sensitized person, leading to tissue damage which occurs on subsequent exposure to the same antigen.

Classification of hypersensitivity: The original Gell and Coombs classification categorizes hypersensitivity reactions into four subtypes according to the type of immune response and the effector mechanism responsible for cell and tissue injury.

1. **Type I (Immediate or IgE mediated)**
2. **Type II (Cytotoxic or IgG/IgM mediated)**
3. **Type III (IgG/IgM immune complex mediated)**
4. **Type IV (Delayed type or T-cell mediated)**

Mechanism of anaphylaxis (Type I):

- **First exposure:**
    1. Antigen enters the body.
    2. Taken up by macrophages.
    3. Macrophages process and present the antigen to helper T cells.
    4. Helper T cell stimulates B cells.
    5. Plasma cell formation and IgE production.
    6. IgE binds to mast cells and basophils, causing sensitization.
- **Re-exposure to the same antigen in an already sensitized person:**
    1. Reaction of antigen with antibody attached to the surface of mast cells and basophils.
    2. Degranulation of mast cells and basophils.
    3. Release of mediators (histamine, serotonin).
    4. This leads to:
        - Vasodilation
        - Increased vascular permeability
        - Increased glandular secretion (asthma, allergic rhinitis)
        - Contraction of smooth muscle.
    5. Anaphylaxis (Anaphylactic shock).

The Process of a Type I Hypersensitivity Reaction:

The diagram and text illustrate the two main stages of this allergic reaction: sensitization and re-exposure.

1. First Exposure (Sensitization):

An Antigen Presenting Cell (APC), such as a macrophage or dendritic cell, takes up an allergen (labeled as Ag).

The APC processes the allergen and presents it to a T helper cell (Th cell), specifically a Th2 cell.

The Th2 cell becomes activated and releases cytokines, specifically Interleukin-4 (IL-4) and Interleukin-5 (IL-5).

These cytokines activate B cells.

The activated B cells proliferate and differentiate into plasma cells.

Plasma cells then produce and secrete Immunoglobulin E (IgE) antibodies.

This IgE antibody has a specific affinity for receptors on the surface of mast cells and basophils, and it binds to them via its Fc region. This process is called sensitization; at this stage, there are no allergy symptoms.

2. Subsequent Exposure (Re-exposure):

Upon a second exposure to the same allergen, the allergen binds to the IgE antibodies that are already attached to the mast cells.

This cross-linking of IgE antibodies triggers the mast cell to release the contents of its granules, a process called degranulation.

These granules contain potent inflammatory mediators such as histamine, leukotrienes, and prostaglandins.

The release of these molecules causes the classic symptoms of an allergic reaction, which can range from mild (like hives and rhinitis) to severe and life-threatening (like anaphylaxis).

Mechanism of Type I hypersensitivity: Antibody responsible: IgE. IgE antibody is induced by an allergen and binds to mast cells and basophils. When exposed to the allergen again, the allergen cross-links the bound IgE, which induces degranulation and release of mediators. Example: Histamine, serotonin.

Mechanism of Type II hypersensitivity: Antibody responsible: IgG. Antigen on the cell surface combines with antibody; this leads to complement-mediated lysis. Example: Transfusion or Rh reactions, Autoimmune hemolytic reactions.

Mechanism of Type III hypersensitivity: Antibody responsible: IgG. Antigen-antibody immune complexes are deposited in tissue; complement is activated and polymorphonuclear cells are attracted to the site. They release lysosomal enzymes, causing tissue damage. Example: Acute glomerulonephritis.

Mechanism of Type IV hypersensitivity: Helper T lymphocytes sensitized by an antigen release lymphokines upon second contact with the same antigen. The lymphokines induce inflammation and activate macrophages, which in turn release various mediators. Example: Tuberculin test, Contact dermatitis.

Q9. Short note on opsonization.

Opsonization: is the attachment of antigen molecules to the surface of phagocytes (macrophages, neutrophils, NK cells, mast cells, monocytes, dendritic cells) by the help of some special molecules, e.g., Immunoglobulin and complement.

Importance:

1. Opsonization results in much more efficient phagocytosis.
2. Opsonization is especially important against microorganisms with antiphagocytic structures such as capsules.

Opsonization occurs by either of two reactions:

1. The Fc portion of IgG interacts with its receptors on the phagocyte surface to facilitate ingestion.
2. IgG or IgM activates complement to yield C3b, which interacts with its receptor on the surface of the phagocyte.

Q10. Short note on ASO titre.

ASO titre means antibody against streptolysin O. It is mainly produced 3-4 weeks after a recent streptococcal sore throat infection.
Function of ASO: ASO blocks hemolysis by streptolysin-O (the main function of streptolysin-O is to cause hemolysis).

Clinical importance:

1. Important in the diagnosis of rheumatic fever or scarlet fever.
2. In throat infection, the ASO titre rises, but in skin infection, the ASO titre does not rise because the antibody against streptolysin-O is destroyed by skin lipase.
3. ASO titre indicates a past or present infection with streptococci.

Normal value: The normal level of ASO titre is 160-200 IU/ml.
Result: ASO titre higher than normal value (160-200 IU/ml) indicates a recent infection with streptococci.
Interpretation: Some non-streptococcal conditions that occasionally cause an increase in ASO titre are:

- (i) Pneumococcal pneumonia
- (ii) Tuberculosis
- (iii) Gonorrhea
- (iv) Hepatitis

Q11. Short note on T lymphocyte (T-cell).

T cells are one type of WBC. They mature and differentiate in the thymus. Due to thymus dependency, they are called T-cells.

- **Origin:** Derived from bone marrow stem cells.
- **Maturation:** Mature and differentiate in the thymus.

Classification:

1. **Helper T cell (CD4):** Recognizes antigen through class-II MHC protein.
    - (i) Helper T cell-1 (TH-1)
    - (ii) Helper T cell-2 (TH-2)
2.
3. **Cytotoxic T cell (CD8):** Recognizes antigen through class-I MHC protein.

Surface receptor:

1. **Antigen receptor:** The antigen receptor of a T cell is known as the T-cell receptor (TCR).
2. **Other receptor:** Receptor for sheep RBC.

Distribution:

1. Peripheral blood
2. Lymphoid tissue:
    - (a) Paracortical area of the lymph node.
    - (b) Periarteriolar sheath of the spleen.
3.

Important function: Two main categories:

1. **Regulatory function (Helper T-cell or CD4)**
2. **Effective function (Cytotoxic T-cell or CD8)**

- **Regulatory function:** These functions are primarily mediated by helper T cells (CD4), which produce interleukins (IL).
    - (i) IL-4 and IL-5, which help B-cells to produce antibodies.
    - (ii) IL-2, which activates CD4 and CD8 T-cells.
    - (iii) Gamma interferon, which activates macrophages.

- **Effective function:** These are carried out primarily by cytotoxic T-cells (CD8), which kill various infected cells and tumor cells and allografts. They kill by releasing perforins and granzymes.

Q12. List the primary and secondary lymphoid organs. Write down the definition and causes of immune suppression. Define immune response. Give the difference between primary and secondary immune response. Compare and contrast between B & T lymphocytes.

List of the lymphoid organs:

1. **Primary lymphoid organs:**

    (a) Thymus (T-cell)

    (b) Bone marrow (B-cell)

2. **Secondary lymphoid organs:**

    (a) Lymph node

    (b) Spleen

    (c) Mucosal-associated lymphoid tissue (MALT), like tonsil, appendix.

Immune Suppression

Definition:

Immunosuppression is a state in which the immune system's ability to mount an effective response to antigenic stimulation is diminished or absent.

Causes:

1. **Primary / Congenital immunodeficiency disorders** – The immunodeficient conditions that result before birth are called primary immunodeficiencies.

   a) **B-cell deficiencies:**
      - X-linked agammaglobulinemia
      - Selective IgA/IgG/IgM deficiencies

   b) **T-cell deficiencies:**
      - Thymic aplasia (DiGeorge syndrome)
      - Chronic mucocutaneous candidiasis
      - Hyper-IgM syndrome.

   c) **Combined B and T cell deficiencies:**
      - Severe combined immunodeficiency (SCID)
      - Wiskott–Aldrich syndrome
      - Ataxia-telangiectasia.

   d) **Complement deficiencies:** Hereditary angioedema
      - Recurrent infections
      - Autoimmune diseases
      - paroxysmal nocturnal hemoglobinuria

   e) **Phagocytic deficiencies:**
      - Chronic granulomatous disease
      - Chediak–Higashi syndrome
      - Job's (Hyper-IgE) syndrome
      - Leukocyte adhesion deficiency
      - Neutropenia
      - MPO deficiency.

2. **Secondary / Acquired immunodeficiency disorders** – The immunodeficient conditions that result after birth are called secondary immunodeficiencies.

   a) **B-cell deficiencies:**

      - Common variable hypogammaglobulinemia
      - Malnutrition.

   b) **T-cell deficiencies:**
      - AIDS
      - Measles.

c) **Complement deficiencies:**
   - Liver failure
   - Malnutrition.

d) **Phagocytic deficiencies:**
   - Neutropenia.

Ref. Endeavour Microbiology and immunology 17th

Immune Response

Definition:
The immune response means any defensive reaction to foreign material by the immune system

Primary vs. Secondary Immune Response

Feature	Primary Immune Response	Secondary Immune Response
Antigen exposure	First encounter with antigen	Subsequent encounters with the same antigen
Lag phase	7–10 days	1–3 days
Antibody isotype	Predominantly IgM, followed by IgG	Predominantly IgG (higher titer), also IgA or IgE depending on antigen
Peak response	Low	Much higher
Duration	Short-lived	More sustained
Antibody affinity	Lower (few somatic mutations)	Higher due to affinity maturation
Memory	Absent initially	Memory B and T cells rapidly activated

(Ref: Abbas & Lichtman; Kuby Immunology)

Comparison of B and T Lymphocytes

Feature	B Lymphocytes	T Lymphocytes
Site of maturation	Bone marrow	Thymus
Participation in humoral immunity	Directly	Indirectly (only CD4+T-cells)
Participation in Cell mediated immunity	No	Yes
Presence of IgM on surface	Present	Absent
CD molecules	CD14,CD20,CD22	CD3 for all T-cella CD4 for helper T-cell CD8 for cytotoxic T-cell
Antigen receptor	IgM and IgD	T-cells receptors
Production of immunoglobulin	After differentiation to plasma cells B-cells can produce immunoglobulins	They cannot produce immunoglobulins
Function as antigen presenting cell	Yes	No

Ref. Endeavour Microbiology and immunology 17th

Q-13. Define immunoglobulin. Give the properties of different immunoglobulins in tabulated form with classification and the difference between IgG & IgM.

Immunoglobulin: Immunoglobulins are proteins of animal origin which have structural similarities with antibodies, but with or without antibody like activity.

There is no structural difference between antibodies and immunoglobulins but there are functional differences.

Or

Immunoglobulins (Ig) are glycoproteins produced by plasma cells in response to antigenic stimulation and function as antibodies. They play a central role in the recognition, neutralization, and elimination of antigens.

Classification of Immunoglobulins (Iso-types):

Immunoglobulins are classified on the basis of the structure of the constant region. So, on the basis of amino acid sequence in the constant region of heavy chain, the immunoglobulins are divided into 5 classes.

1. **IgG (80%) - contains γ chain:** On the basis of minor difference in amino acid sequence in the constant region of heavy chain the IgG are of 4 sub-classes:
    a. IgG_1
    b. IgG_2
    c. IgG_3
    d. IgG_4

2. **IgA (13%) - contains α chain:** On the basis of minor difference in amino acid sequence in the constant region of heavy chain the IgA are of 2 sub-classes:
    a. IgA_1 (93%)
    b. IgA_2 (7%)

3. **IgM (06%) - contains μ chain.**
4. **IgD - contains δ chain.**
5. **IgE - contains ε chain.**

Ref. Endeavour Microbiology and immunology 17th

Properties of Immunoglobulins

Class	Structure	Serum concentration	Half-life	Main functions
IgG	Monomer	~75% of total Ig	23 days	Crosses placenta (provides passive immunity to fetus), opsonization, complement activation, neutralization of toxins & viruses
IgA	Monomer in serum, dimer in secretions	~15%	6 days	Found in secretions (saliva, tears, colostrum, mucosal surfaces); protects mucosal surfaces by preventing microbial adherence

IgM	Pentamer (with J chain)	~10%	5 days	First antibody produced in primary response, excellent complement activator, effective in agglutination
IgE	Monomer	<1%	2 days	Mediates type-I hypersensitivity (allergy, anaphylaxis), defense against parasites (helminths)
IgD	Monomer	<1%	3 days	Functions mainly as antigen receptor on immature B-cells; role in B-cell activation

Abbas AK, Lichtman AH. Cellular and Molecular Immunology, 10th ed.
Kuby J. Kuby Immunology, 8th ed.
Roitt IM. Roitt's Essential Immunology, 13th ed.

Differences between IgG and IgM

Feature	IgG	IgM
Structure	Monomer	Pentamer
Abundance	Most abundant in serum (80%)	6% of serum Ig
Distribution	Equally distributed between intravascular and extravascular spaces	Distributed for predominantly in the intravascular spaces
Placental transfer	Crosses placenta → passive immunity to fetus	Does not cross placenta
Agglutination	Moderate	Strong (due to pentameric structure)
Antigen receptor on B cell	Absent	Present
Half-life	Longest (23 days)	Short (5 days)
Immune response	Dominant in secondary immune response	First antibody in primary immune response

Ref. Endeavour Microbiology and immunology 17th

Q-14. What is complement? Describe the complement activation pathway.

Complement: complement is a collective term used to designate a group of plasma proteins at least 20 in number, which normally remain inactive form and when activated they act in concert in an orderly sequence to exert their biological effects.

Ref. Endeavour - Prof. Akram 5th

Or

The complement system is a group of plasma proteins and membrane-associated proteins that act as part of the innate and adaptive immune system.

Abbas AK, Lichtman AH. Cellular and Molecular Immunology, 10th ed.

Complement Activation Pathways

Complement can be activated by **three major pathways**, all converging at the activation of **C3** and leading to formation of the **membrane attack complex (MAC, C5b–C9)**.

1. Classical pathway:

IgG & IgM activates this pathway.

a) C1 binds Fc fragments of IgG or IgM then C1 is activated.
b) Activated C1 converts C4 into C4a & C4b. also C2 into C2a & C2b.
c) C4bC2b remains in the reaction and is called C3 convertase.
d) C4b,C2b convert C3 into C3a & C3b.
e) C4b,C2b,C3b are called C5 convertase. This converts C5 into C5a & C5b.
f) C6, C7 then bind to C5b,6,7.
g) C8 & C9 bind with C5b,6,7 to produce C5b,6,7,8,9. This is called the membrane attack complex (MAC).

2. Alternate pathway:

Some case IgA activates complement in this pathway.
Moreover the activators of this pathway includes -
i. Lipopolysaccharide of Gram negative bacteria (endotoxin).
ii. Teichoic acid of gram positive bacteria.
iii. viral envelope.
iv. Fungal cell wall.
v. Cobra venom.
vi. Aggregated IgA.

Steps of activation:
a) the activators presence of D & Mg++ convert small portion of C3 into C3a & C3b.
b) Again the activation presence of D & Mg++ convert Factor B into Ba & Bb.
c) C3b,Bb is active enzyme and with a protein called C3 convertase in alternate pathway. C3b,Bb convert more C3 into C3a & C3b.

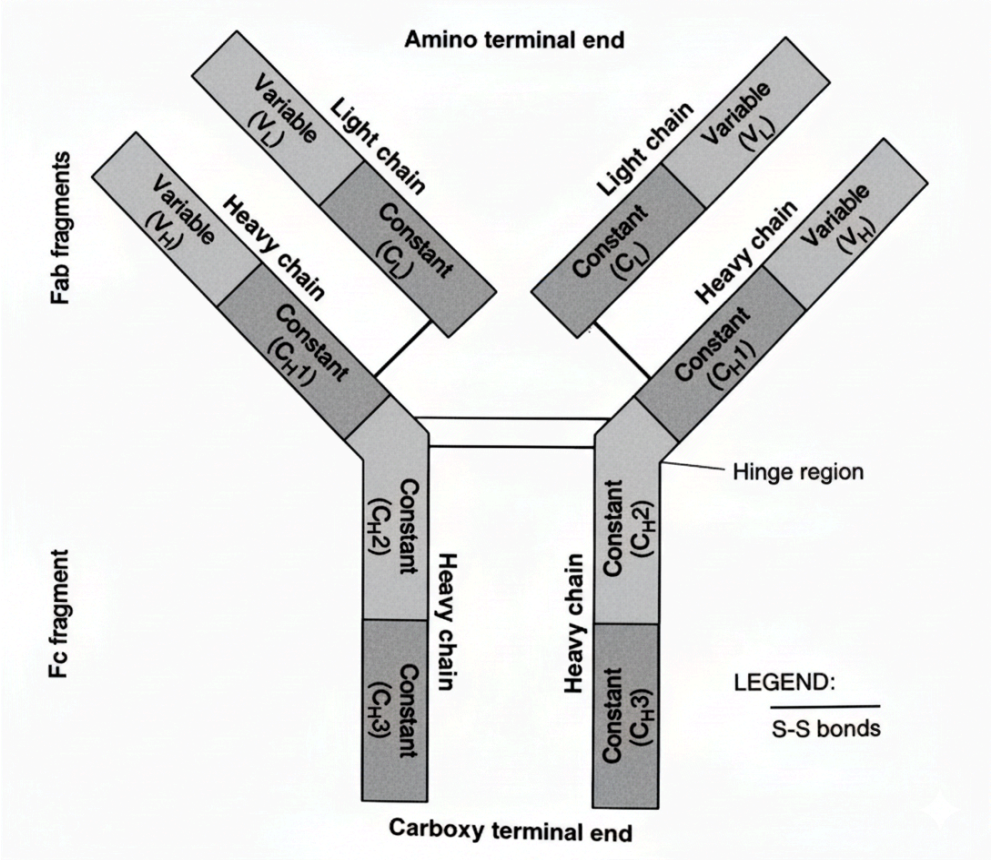
d) $(C3b)_2Bb$ is called C5 convertase and convert C5 into C5a & C5b.
Rest of the pathway are same as classical pathway.

3. Lectin pathway:

Activators: Streptococcus pneumoniae, Klebsiella, Haemophilus influenzae.
a) Microbial surface bind with mannan-binding lectin.
b) This then activa protease and cleaves C4 & C2.
c) Rest of the reactions of this pathway are same as classical pathway.

Q-15. Write a short note on herd immunity and PCR.

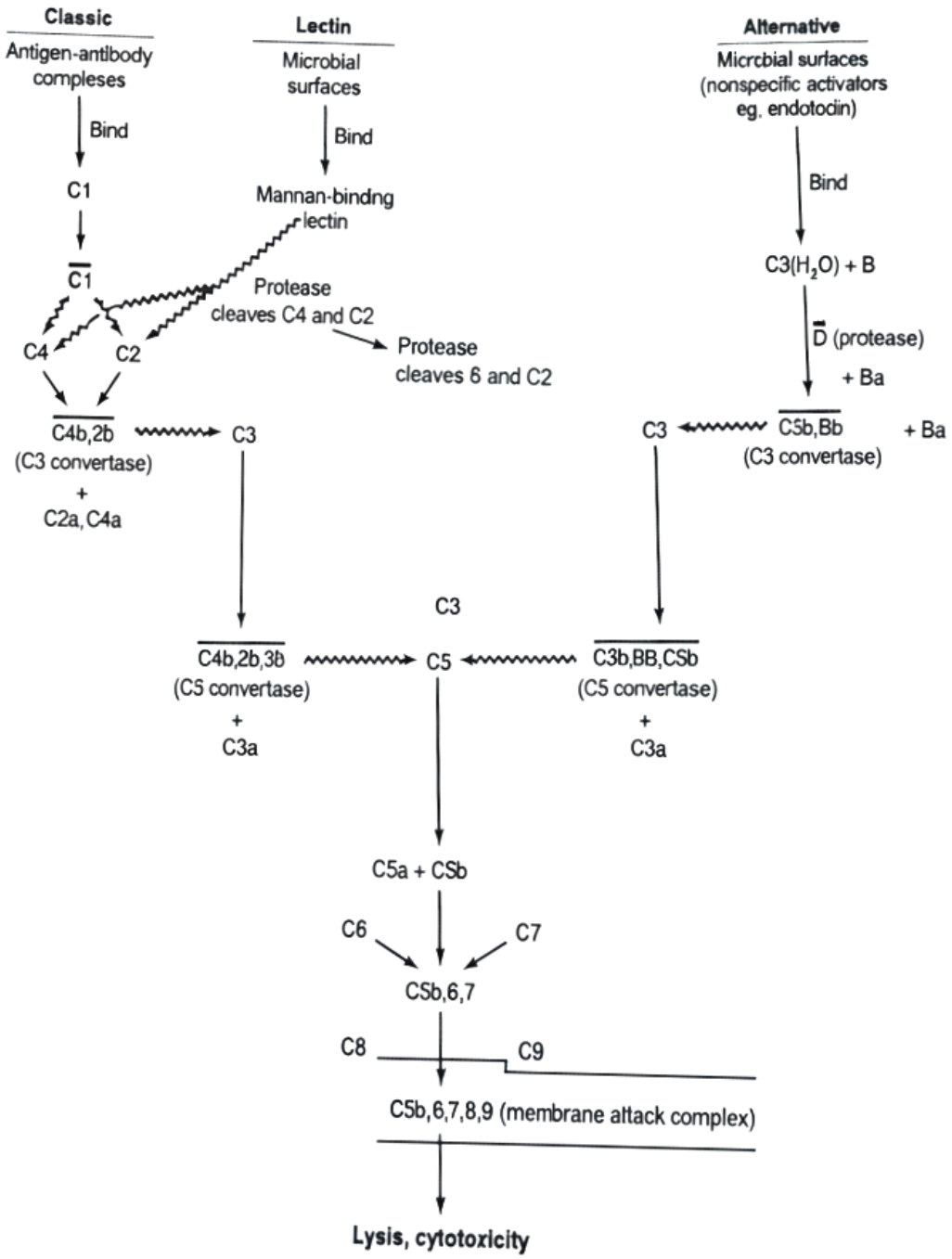
Q-16. Draw and label the structure of an immunoglobulin molecule.



Q-17. Diagram the complement activation pathway. What is the biological function of complement?

Complement can be activated by **three major pathways**, all converging at the activation of **C3** and leading to formation of the **membrane attack complex (MAC, C5b–C9)**.

1. Classical pathway
2. Alternative pathway
3. Lectin pathway



1. **Inflammation:** Mainly by C5a, and to a lesser extent by C3a and C4a.

2. **Chemotaxis (C3a & C5a):** They have chemotactic activity.

3. **Opsonization (C3b):** C3b helps in opsonization (which helps in phagocytosis).

4. **Cell lysis / cytolysis:** Insertion of the C5b,6,7,8,9 complex (membrane attack complex) into the cell membrane leads to killing or lysis of many types of cells including RBC, bacteria, enveloped virus, malaria parasite and tumor cells.

5. **Activation of B lymphocyte and stimulation of antibody production:** C3b binds and stimulates B-lymphocytes, thus causing antibody production.

6. **Removing harmful immune complexes from the body:** C3b and to a lesser extent, C4b help to remove harmful immune complexes from the body.

7. **Anaphylatoxin (C3a, C4a and C5a):** They cause degranulation of mast cells with the release of mediators. i.e. histamine, that causes anaphylaxis (skin rash, hypotension, bronchoconstriction, diarrhoea & shock).

8. **Immune adherence:** C3b is responsible for immune adherence.

9. **Virus neutralization (C1 & C4):** They play role in neutralization of viruses.

[Ref- Endeavour/ Prof. Akram / 5th / 149,150]

Q-18. Write down the classification of immune responses. Differentiate between humoral and cell-mediated immune responses.

Types of immune response: There are two types of immune response:-

1. Primary immune response
2. Secondary immune response

1. Primary immune response: Primary immune response is the immune response that occurs on first exposure to any antigen.

Primary immune response is characterized by:

a) Long lag phase (latent period)
b) IgM is the predominant antibody
c) Low titer of antibody with low quality and short duration.

2. Secondary immune response: Secondary immune response is one that occurs when the second time (or more) an antigen enters the body.

Secondary immune response is characterized by:

a) Short lag phase (latent period)
b) IgG is the predominant antibody.
c) High titer of antibody with high quality and long duration.
d) Memory cells are present.

Difference between Humoral and Cell-Mediated Immune Responses

Feature	Humoral Immune Response	Cell-Mediated Immune Response
Definition	Immunity mediated by B lymphocytes and antibodies in body fluids	Immunity mediated by T lymphocytes acting directly on infected or abnormal cells
Main Cells	B cells → Plasma cells	T cells ($CD4^+$ helper, $CD8^+$ cytotoxic), macrophages
Effector Molecules	Antibodies (IgG, IgM, IgA, IgE, IgD)	Cytokines, perforin, granzymes
Targets	Extracellular pathogens (bacteria, fungi, parasites, toxins, viruses before entry into cells)	Intracellular pathogens (viruses, some bacteria & protozoa), tumor cells, grafts
Mechanism of Action	Neutralization, opsonization, complement activation, agglutination, lysis	Direct killing of infected cells, activation of macrophages, regulation via cytokines
Memory	Memory B cells	Memory T cells
Deficiency → Susceptibility	Extracellular bacterial infections (e.g., *Streptococcus pneumoniae*)	Viral, fungal, intracellular bacterial infections (e.g., TB, herpes, Candida)

Abbas & Lichtman: *Cellular and Molecular Immunology, 10th Ed.*
Janeway et al.: *Janeway's Immunobiology, 9th Ed.*

Q-19. Give examples of agglutination and precipitation.

Agglutination

1. **Widal test** – detection of *Salmonella* antibodies (typhoid fever).
2. **Weil–Felix test** – diagnosis of rickettsial infections.
3. **Blood grouping (ABO, Rh typing)** – antigen–antibody reaction on RBC surface.
4. **Latex agglutination test** – detection of CRP, rheumatoid factor, bacterial antigens.

Precipitation

1. **Radial immunodiffusion (Mancini method)** – quantitation of immunoglobulins.
2. **Ouchterlony double diffusion** – antigen–antibody diffusion in agar gel, used to detect antigenic relationship.
3. **Ring test** – e.g., Ascoli's thermoprecipitation test for anthrax.
4. **VDRL (Venereal Disease Research Laboratory test)** – uses flocculation (a type of precipitation) for syphilis screening

Abbas & Lichtman: *Cellular and Molecular Immunology*.
Roitt's *Essential Immunology*.

Q-20. Write a short note on radioimmunoassay and MHC.

Q-21. Write a short note on the Widal test.

Q-22. Give the properties of IgM, IgA, and IgE.

Q-23. Write a short note on tumor antigens.

Q-24. Write down the comparison between Class I and Class II MHC.

Q-25. Define allergy. Explain Type I hypersensitivity with an example.

Q-26. Write a short note on phagocytosis.

Q-27. Write about CFT (Complement Fixation Test). Mention its use.

Q-28. Principles and interpretation of the Coombs test.

Q-29. Short note on ICT (Immunochromatographic Test).

Q-30. Describe the pathogenesis of anaphylaxis, including the type of antibody, the cells, and mediators involved.

Q-31. What are the causes of secondary immune deficiency?

Q-32. What are the determinants of innate immunity?

Q-33. Classify different types of serological tests performed in the laboratory with their examples. List them properly formatted.